

Yue Zhao

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Employment

- **Michigan State University** 08/2023 - now
Postdoctoral Researcher
– Mentor: Prof. Huan Lei

Education

- **Shanghai Jiao Tong University** 09/2019 – 06/2023
Ph.D. in Mathematics
– Advisor: Prof. Zhenli Xu
– Co-advisor: Prof. Xiaoping Wang
- **Hong Kong University of Science and Technology** 02/2022 - 01/2023
Visiting Scholar
- **Shanghai Jiao Tong University** 09/2015 - 06/2019
B.S. in Mathematics

Research Interests

- Multiscale modeling for structure-preserving machine learning models;
- Fast algorithms in molecular dynamics and the Monte Carlo method;
- Numerical methods for partial differential equations;
- Hydrodynamics and interfacial problems;
- Data assimilation and Bayesian inverse problems.

Publications (Google Scholar)

* Corresponding author; [†] Co-first author; other authors listed a-z.

Peer-Reviewed Journal Articles

1. **Yue Zhao**[†], Guosheng Fu and Huan Lei, From molecular dynamics to kinetic models: data-driven generalized collision operators in 1D3V plasmas, *arXiv:2603.27828*, 2026.
2. **Yue Zhao**[†] and Huan Lei, Fast spectral separation method for kinetic equation with anisotropic non-stationary collision operator retaining micro-model fidelity. *J. Comput. Phys.*, in press, *arXiv:2510.15093*, 115205, 2026.
3. **Yue Zhao**[†], Joshua W. Burby, Andrew J. Christlieb and Huan Lei*, Data-driven construction of a generalized kinetic collision operator from molecular dynamics. *Phys. Rev. Lett.*, 135(18):185101, 2025.
4. Zhenli Xu, **Yue Zhao** and Qi Zhou, Variance-reduced random batch Langevin dynamics. *J. Chem. Phys.*, 161(24):244110, 2024.
5. Jiuyang Liang, Zhenli Xu and **Yue Zhao**, Energy stable scheme for random batch molecular dynamics. *J. Chem. Phys.*, 160(3):034101, 2024.

6. Jiuyang Liang, Zhenli Xu and **Yue Zhao**, Improved random batch Ewald method in molecular dynamics simulations. *J. Phys. Chem. A*, 126(22):3583-3593, 2022.
7. Jiuyang Liang[†], Pan Tan[†], **Yue Zhao**, Lei Li, Shi Jin, Liang Hong* and Zhenli Xu*, Superscalability of the random batch Ewald method. *J. Chem. Phys.*, 156(1):014114, 2022.
8. Jiuyang Liang, Zhenli Xu and **Yue Zhao**, Random-batch list algorithm for short-range molecular dynamics simulations *J. Chem. Phys.*, 155(4):044108, 2021.
9. Shi Jin, Lei Li, Zhenli Xu and **Yue Zhao**, A random batch Ewald method for particle systems with Coulomb interactions. *SIAM J. Sci. Comput.*, 43(4):B937-B960, 2021.
10. Lei Li, Zhenli Xu, and **Yue Zhao**, A random-batch Monte Carlo method for many-body systems with singular kernels. *SIAM J. Sci. Comput.*, 42(3):A1486-A1509, 2020.

Work in Preparation

11. **Yue Zhao**[†], Andrew J. Christlieb and Huan Lei, Data-driven generalized collision operators for multi-species kinetic systems, preprint, 2026.
12. **Yue Zhao**[†], Shilin Dou and Huan Lei, A scalable fast particle method for the Landau collision operator, preprint, 2026.

Awards

• AMS-Simons Travel Award AMS and Simons Foundation.	2026-28
• MWNADay Travel Award	2026
• Shanghai Outstanding Doctoral Graduates	2022
• Wenjun Wu Scholarship Shanghai Jiao Tong University	2021
• Weichai Power Scholarship Shanghai Jiao Tong University	2021
• Second Prize of Zhiyuan Honored Ph.D Students Academic Forum Shanghai Jiao Tong University	2021
• First Prize in the 8th National College Student Mathematics Competition Second Prize in the Final Round Chinese Mathematical Society	2017
• Excellent Students Scholarship in University Shanghai Jiao Tong University	2015-2019

Academic Service

Journal Reviewer:

- Advances in Applied Mathematics and Mechanics
- IEEE Transactions on Parallel and Distributed Systems
- IEEE Transactions on Computers
- IEEE Transactions on Industrial Electronics

Conference Reviewer:

- ICLR 2025 Workshop on Machine Learning for Molecules and Materials

Talks

- **SIAM MDS 26** 11/2026
"Structure-Preserving Construction of Collision Operators for Kinetic Equations from Molecular Dynamics"
Salt Palace Convention Center, Salt Lake City, UT, USA.
- **SIAM Annual Meeting 2026** 07/2026
"Structure-Preserving Construction of Collision Operators for Kinetic Equations from Molecular Dynamics"
Huntington Convention Center of Cleveland, Cleveland, OH, USA.
- **Midwest Numerical Analysis Day** 04/2026
"Structure-Preserving Construction of Collision Operators for Kinetic Equations from Molecular Dynamics"
University of Notre Dame, Notre Dame, IN, USA.
- **The Great Lakes section of SIAM (GLSIAM)** 09/2025
"Data-driven Learning and Fast Computation of the Kinetic Collision Operator from Molecular Dynamics"
Illinois Institute of Technology, Chicago, IL, USA.
- **2024 Joint Mathematics Meetings** 01/2024
"Random-Batch Ewald Method"
San Francisco, CA, USA.
- **CSIAM 2021 Annual Meeting (Student Forum)** 10/2021
"Random-Batch Ewald Method"
Hefei, China.
- **Zhiyuan Honored Ph.D Students Academic Forum (Second Prize)** 07/2021
"Random-Batch Ewald Method"
Shanghai Jiao Tong University, Shanghai, China.

Conference and Workshop Presentations

- **NSF Computational Math Meeting** 05/2025
"Data-Driven Construction of the Kinetic Collision Operator from Molecular Dynamics" (Poster)
The University of Utah, Salt Lake City, UT, USA.
- **CHARMNET meeting** 12/2024
Virginia Tech, Blacksburg, VA, USA.
- **CHARMNET meeting** 12/2023
"Learning Collision Operator from Micro-Scale Descriptions" (Poster)
Michigan State University, East Lansing, MI, USA.
- **Mathematics in Action (MiA2021)** 06/2021
"Random-Batch Ewald Method" (Poster)
Duke Kunshan University, Kunshan, Jiangsu, China.
- **The Forum on Mathematics and Industrial Innovations** 03/2020
Shanghai Science Hall, Shanghai, China.
- **Annual Meeting of Shanghai Society for Industrial and Applied Mathematics** 11/2019
Fudan University, Shanghai, China.
- **15th Shanghai Symposium on Scientific and Engineering Computing Methods** 11/2019
Shanghai Maritime University, Shanghai, China.

Teaching experience

Michigan State University

- **STT 351H, Probability and Statistics for Engineering** Fall 2025

Shanghai Jiao Tong University

- **Teaching Assistant**, Numerical Analysis and Programming
- **Teaching Assistant**, Numerical Analysis and Programming
- **Teaching Assistant**, Mathematical Analysis I

Spring 2021

Spring 2020

Fall 2019

NYU Shanghai

- **Teaching Assistant**, Math for Economics I and II

Fall 2021